Transfer: thermal energy

antocciphysics | TEACHER KEY | CPA

Learning target: Choose an energy-transfer model and communicate its limits.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 5, Day 9; 3 course days later.*
30% efficiency; 350 J rejected, assuming a complete cycle.

02. Explain a worked example (worked)
120 in=50 work out +70 stored.

03. Look and explain (model)
Flow from block to water; final common temperature lies between initial values under normal mixing assumptions; exact value needs heat capacities.

04. Build the solution (practice)
800 J warming; 600 J phase change.

05. Use the idea (reason)
350 J; efficiency 30%, must also satisfy second-law bound for reservoir temperatures.

06. Choose a method (mixed) *Review: Unit 4, Day 22; 12 course days later.*
6 W then 3 W. Same energy over twice the time gives half the power.

07. Exit check (exit)
+25 J.

08. STRETCH: Apply it another way (stretch)
Example work 20 J out requires heat 90 J in; infinitely many pairs share ΔU 70.

Teaching note: Self-contained daily transfer worksheet. Keep any separately assigned formal assessment independent.

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01. Bring back an earlier idea (retrieve) *Review: Unit 5, Day 9; 3 course days later.*
Work 720 J; efficiency 30%.

02. Explain a worked example (worked)
120 in=50 work out +70 stored.

03. Represent and interpret (model)
Flow from block to water; final common temperature lies between initial values under normal mixing assumptions; exact value needs heat capacities.

04. Solve and justify (practice)
33, 400+4200=37, 600 J.

05. Analyze the reasoning (reason)
350 J; efficiency 30%, must also satisfy second-law bound for reservoir temperatures.

06. Choose a method (mixed) *Review: Unit 4, Day 22; 12 course days later.*
v_{escape}=12 m/s. Assumptions include ideal central gravity, no atmosphere, no further thrust and negligible other bodies.

07. Exit check (exit)
Max efficiency 40%; max work 400 J; irreversibility/losses reduce actual work.

08. STRETCH: Apply it another way (stretch)
ΔU 400 J; Q 460 J. No phase change, suitable constant heat capacity and specified system accounting.

Teaching note: Self-contained daily transfer worksheet. Keep any separately assigned formal assessment independent.